

1. Общая выручка по категориям

```
SELECT "Category",
       ROUND(SUM("Sales"), 2) AS total_sales
FROM train
GROUP BY "Category"
ORDER BY total_sales DESC;
```

query	
Category	total_sales
Technology	827 455,87
Furniture	728 658,58
Office Supplies	705 422,33

2. Топ-10 товаров по выручке

```
SELECT "Product Name",
       ROUND(SUM("Sales"), 2) AS total_sales
FROM train
GROUP BY "Product Name"
ORDER BY total_sales DESC
LIMIT 10;
```

query-2	
Product Name	total_sales
Canon imageCLASS 2200 Advanced Copier	61 599,82
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27 453,38
Cisco TelePresence System EX90 Videoconferencing Unit	22 638,48
HON 5400 Series Task Chairs for Big and Tall	21 870,58
GBC DocuBind TL300 Electric Binding System	19 823,48
GBC Ibimaster 500 Manual ProClick Binding System	19 024,5
Hewlett Packard LaserJet 3310 Copier	18 839,69
HP Designjet T520 Inkjet Large Format Printer - 24" Color	18 374,9
GBC DocuBind P400 Electric Binding System	17 965,07
High Speed Automatic Electric Letter Opener	17 030,31

4. Сравнение продаж с предыдущим месяцем (оконная функция LAG)

```
WITH monthly AS (
    SELECT DATE_TRUNC('month', "Order Date") AS month,
           ROUND(SUM("Sales"), 2) AS sales
    FROM train
    GROUP BY month
)
SELECT month,
       sales,
       LAG(sales, 1) OVER (ORDER BY month) AS
prev_month_sales,
       ROUND((sales - LAG(sales, 1) OVER (ORDER BY month))
/
           NULLIF(LAG(sales, 1) OVER (ORDER BY month),
0) * 100, 2) AS growth_pct
FROM monthly
ORDER BY month;
```

query-4			
month	sales	prev_month_sales	growth_pct
2015-01-01 00:00:00	14 205,71		
2015-02-01 00:00:00	4 519,89	14 205,71	'-68,18
2015-03-01 00:00:00	55 205,8	4 519,89	1 121,4
2015-04-01 00:00:00	27 906,85	55 205,8	'-49,45
2015-05-01 00:00:00	23 644,3	27 906,85	'-15,27

3. Динамика продаж по месяцам

```
SELECT DATE_TRUNC('month', "Order Date") AS
month,
       COUNT(DISTINCT "Order ID") AS
number_of_orders,
       ROUND(SUM("Sales"), 2) AS monthly_sales
FROM train
GROUP BY month
ORDER BY month;
```

query-3		
month	number_of_orders	monthly_sales
2015-01-01 00:00:00	30	14 205,71
2015-02-01 00:00:00	28	4 519,89
2015-03-01 00:00:00	69	55 205,8
2015-04-01 00:00:00	63	27 906,85
2015-05-01 00:00:00	68	23 644,3
2015-06-01 00:00:00	64	34 322,94
2015-07-01 00:00:00	64	33 781,54
2015-08-01 00:00:00	70	27 117,54
2015-09-01 00:00:00	129	81 623,53
2015-10-01 00:00:00	78	31 453,39
2015-11-01 00:00:00	145	77 907,66
2015-12-01 00:00:00	139	68 167,06
2016-01-01 00:00:00	29	18 066,96
2016-02-01 00:00:00	36	11 951,41
2016-03-01 00:00:00	75	32 339,32
2016-04-01 00:00:00	71	34 154,47
2016-05-01 00:00:00	72	29 959,53
2016-06-01 00:00:00	66	23 599,37
2016-07-01 00:00:00	64	28 608,26
2016-08-01 00:00:00	67	36 818,34
2016-09-01 00:00:00	137	63 133,61
2016-10-01 00:00:00	86	31 011,74
2016-11-01 00:00:00	157	75 249,4
2016-12-01 00:00:00	159	74 543,6
2017-01-01 00:00:00	48	18 542,49
2017-02-01 00:00:00	45	22 978,82
2017-03-01 00:00:00	85	51 165,06
2017-04-01 00:00:00	88	38 679,77
2017-05-01 00:00:00	107	56 656,91
2017-06-01 00:00:00	95	39 724,49
2017-07-01 00:00:00	95	38 320,78
2017-08-01 00:00:00	89	30 542,2
2017-09-01 00:00:00	186	69 193,39
2017-10-01 00:00:00	103	59 583,03
2017-11-01 00:00:00	182	79 066,5
2017-12-01 00:00:00	172	95 739,12
2018-01-01 00:00:00	67	43 476,47
2018-02-01 00:00:00	52	19 921
2018-03-01 00:00:00	117	58 863,41
2018-04-01 00:00:00	115	35 541,91
2018-05-01 00:00:00	116	43 825,98
2018-06-01 00:00:00	129	48 190,73
2018-07-01 00:00:00	108	44 825,1
2018-08-01 00:00:00	109	62 837,85
2018-09-01 00:00:00	222	86 152,89
2018-10-01 00:00:00	144	77 448,13
2018-11-01 00:00:00	259	117 938,16
2018-12-01 00:00:00	223	83 030,39

2015-06-01 00:00:00	34 322,94	23 644,3	45,16
2015-07-01 00:00:00	33 781,54	34 322,94	'-1,58
2015-08-01 00:00:00	27 117,54	33 781,54	'-19,73
2015-09-01 00:00:00	81 623,53	27 117,54	201
2015-10-01 00:00:00	31 453,39	81 623,53	'-61,47
2015-11-01 00:00:00	77 907,66	31 453,39	147,69
2015-12-01 00:00:00	68 167,06	77 907,66	'-12,5
2016-01-01 00:00:00	18 066,96	68 167,06	'-73,5
2016-02-01 00:00:00	11 951,41	18 066,96	'-33,85
2016-03-01 00:00:00	32 339,32	11 951,41	170,59
2016-04-01 00:00:00	34 154,47	32 339,32	5,61
2016-05-01 00:00:00	29 959,53	34 154,47	'-12,28
2016-06-01 00:00:00	23 599,37	29 959,53	'-21,23
2016-07-01 00:00:00	28 608,26	23 599,37	21,22
2016-08-01 00:00:00	36 818,34	28 608,26	28,7
2016-09-01 00:00:00	63 133,61	36 818,34	71,47
2016-10-01 00:00:00	31 011,74	63 133,61	'-50,88
2016-11-01 00:00:00	75 249,4	31 011,74	142,65
2016-12-01 00:00:00	74 543,6	75 249,4	'-0,94
2017-01-01 00:00:00	18 542,49	74 543,6	'-75,13
2017-02-01 00:00:00	22 978,82	18 542,49	23,93
2017-03-01 00:00:00	51 165,06	22 978,82	122,66
2017-04-01 00:00:00	38 679,77	51 165,06	'-24,4
2017-05-01 00:00:00	56 656,91	38 679,77	46,48
2017-06-01 00:00:00	39 724,49	56 656,91	'-29,89
2017-07-01 00:00:00	38 320,78	39 724,49	'-3,53
2017-08-01 00:00:00	30 542,2	38 320,78	'-20,3
2017-09-01 00:00:00	69 193,39	30 542,2	126,55
2017-10-01 00:00:00	59 583,03	69 193,39	'-13,89
2017-11-01 00:00:00	79 066,5	59 583,03	32,7
2017-12-01 00:00:00	95 739,12	79 066,5	21,09
2018-01-01 00:00:00	43 476,47	95 739,12	'-54,59
2018-02-01 00:00:00	19 921	43 476,47	'-54,18
2018-03-01 00:00:00	58 863,41	19 921	195,48
2018-04-01 00:00:00	35 541,91	58 863,41	'-39,62
2018-05-01 00:00:00	43 825,98	35 541,91	23,31
2018-06-01 00:00:00	48 190,73	43 825,98	9,96
2018-07-01 00:00:00	44 825,1	48 190,73	'-6,98
2018-08-01 00:00:00	62 837,85	44 825,1	40,18
2018-09-01 00:00:00	86 152,89	62 837,85	37,1
2018-10-01 00:00:00	77 448,13	86 152,89	'-10,1
2018-11-01 00:00:00	117 938,16	77 448,13	52,28
2018-12-01 00:00:00	83 030,39	117 938,16	'-29,6

5. Топ–5 клиентов по выручке

```
SELECT "Customer Name",
      ROUND(SUM("Sales"), 2) AS total_sales
FROM train
GROUP BY "Customer Name"
ORDER BY total_sales DESC
LIMIT 5;
```

query-5

Customer Name	total_sales
Sean Miller	25 043,05
Tamara Chand	19 052,22
Raymond Buch	15 117,34
Tom Ashbrook	14 595,62
Adrian Barton	14 473,57

6. Количество заказов по категориям

```
SELECT "Category",
      COUNT(DISTINCT "Order ID") AS num_orders
FROM train
GROUP BY "Category"
ORDER BY num_orders DESC;
```

query-6

Category	num_orders
Office Supplies	3 676
Furniture	1 727
Technology	1 519